

6-17

6-17 A steam power plant receives heat from a furnace at a rate of 280 GJ/h. Heat losses to the surrounding air from the steam as it passes through the pipes and other components are estimated to be about 8 GJ/h. If the waste heat is transferred to the cooling water at a rate of 165 GJ/h, determine (a) net power output and (b) the thermal efficiency of this power plant. *Answers: (a) 29.7 MW, (b) 38.2 percent*

Given:

- $\dot{Q}_H = 280 \text{ GJ/hr}$
- $\dot{Q}_{\text{loss}} = 8 \text{ GJ/hr}$
- $\dot{Q}_L = 165 \text{ GJ/hr}$

Find:

- $\dot{W}_{\text{net}}$
- η_{th}

This is a **heat engine** since it takes in heat to produce work. We use a first law analysis (closed system) to find the net work output:

$$\Delta U = \sum Q - \sum W \implies \dot{U} = \sum \dot{Q} - \sum \dot{W}$$

Since the system is a cycle, $\Delta U = 0 \implies \dot{U} = 0$, so we have

$$(\dot{Q}_H - \dot{Q}_{\text{loss}} - \dot{Q}_L) - \dot{W}_{\text{net}} = 0$$

$$\implies \dot{W}_{\text{net}} = \dot{Q}_H - \dot{Q}_{\text{loss}} - \dot{Q}_L = [280 - 8 - 165 \text{ GJ/hr}] = 107 \text{ GJ/hr}$$

Converting to MW:

$$\dot{W}_{\text{net}} = [107 \text{ GJ/hr}] \cdot \left[\frac{1 \text{ hr}}{3600 \text{ s}} \right] \cdot \left[\frac{10^3 \text{ MJ}}{1 \text{ GJ}} \right] = 29.72 \text{ MJ/s} = \boxed{29.72 \text{ MW}}$$

The thermal efficiency is the ratio of net work output to heat input:

$$\eta_{\text{th}} = \frac{\dot{W}_{\text{net}}}{\dot{Q}_H} = \frac{[107 \text{ GJ/hr}]}{[280 \text{ GJ/hr}]} = \boxed{0.3821}$$

6-57

6-57 Refrigerant-134a enters the condenser of a residential heat pump at 800 kPa and 35°C at a rate of 0.018 kg/s and leaves at 800 kPa as a saturated liquid. If the compressor consumes 1.2 kW of power, determine (a) the COP of the heat pump and (b) the rate of heat absorption from the outside air.

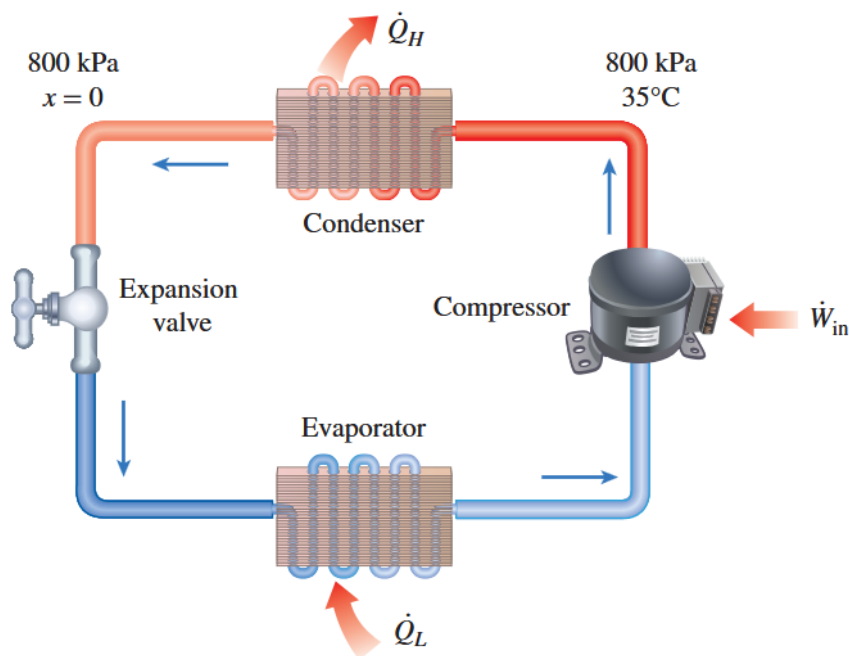


FIGURE P6-57

Given:

Find:

- State 1 (R-134a entering condenser):
 - $P_1 = 800 \text{ kPa}$
 - $T_1 = 35^\circ\text{C}$
 - $\dot{m}_1 = 0.018 \text{ kg/s}$
- State 2 (leaving condenser):
 - $P_2 = 800 \text{ kPa}$
 - Saturated liquid $\Rightarrow x_2 = 0$
- $\dot{W}_{\text{in}} = 1.2 \text{ kW}$
- Coefficient of performance (COP) of the *heat pump* (using work to heat the surroundings)
- Rate of heat absorption from the surroundings ($\dot{Q}_L$)

(a) First, using a first law analysis on the compressor (open system) to determine $\dot{Q}_H$:

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left[h_i + \frac{1}{2}v_i^2 + gz_i \right] - \sum \dot{m}_e \left[h_e + \frac{1}{2}v_e^2 + gz_e \right]$$

We can simplify the first law equation as follows:

- No work is done by the system ($\dot{W} = 0$)
- Steady state ($\frac{dE}{dt} = 0$)
- Neglect potential energy changes ($gz_i \approx gz_e$)
- Neglect kinetic energy changes ($\frac{1}{2}v_i^2 \approx \frac{1}{2}v_e^2$)
- By conservation of mass, $\dot{m}_1 = \dot{m}_2$

Thus, the first law simplifies to:

$$0 = \dot{Q} + \dot{m}_1 h_1 - \dot{m}_2 h_2$$

$$\implies \dot{Q}_H = \dot{m}_1 (h_2 - h_1)$$

From the property tables for R-134a, we have

- State 1: (Superheated vapor) $T_1 > T_{\text{sat}}$ at $P_1 = 800$ kPa. Thus, linear interpolation from the superheated vapor table using values at
 - $T = T_{\text{sat}} = 31.31^\circ\text{C}$: $h_{\text{sat}} = 267.34$ kJ/kg
 - $T = 40^\circ\text{C}$: $h = 276.46$ kJ/kg

$$h - h_{\text{sat}} = \frac{h_{40} - h_{\text{sat}}}{T_{40} - T_{\text{sat}}}(T - T_{\text{sat}})$$

$$h = 267.34 + 1.049(T - 31.31) \implies h_1 = 271.212 \text{ kJ/kg}$$

- State 2 (saturated liquid): $h_2 = h_f = 95.48$ kJ/kg

Thus, we have

$$\dot{Q}_H = [0.018 \text{ kg/s}] \cdot [95.48 - 271.212 \text{ kJ/kg}] = -3.163 \text{ kW}$$

The negative sign confirms that heat is leaving the system. For the COP, we will use the magnitude of $\dot{Q}_H$:

$$\beta_{\text{HP}} = \frac{|\dot{Q}_H|}{\dot{W}_{\text{in}}} = \frac{[3.163 \text{ kW}]}{[1.2 \text{ kW}]} = \boxed{2.635}$$

(b) Applying the first law to the entire heat pump cycle (closed system):

$$\Delta U = \sum Q - \sum W \implies \dot{U} = \sum \dot{Q} - \sum \dot{W}$$

Since the system is a cycle, $\Delta U = 0 \implies \dot{U} = 0$, so we have:

$$(\dot{Q}_H - \dot{Q}_L) - \dot{W}_{\text{in}} = 0$$

Again, using magnitudes for the heat transfer rates (since direction is already accounted for):

$$\implies \dot{Q}_L = \dot{Q}_H - \dot{W}_{\text{in}} = [3.163 - 1.2 \text{ kW}] = \boxed{1.963 \text{ kW}}$$

6-84E

6-84E A heat engine is operating on a Carnot cycle and has a thermal efficiency of 75 percent. The waste heat from this engine is rejected to a nearby lake at 60°F at a rate of 800 Btu/min. Determine (a) the power output of the engine and (b) the temperature of the source. *Answers: (a) 56.6 hp, (b) 2080 R*

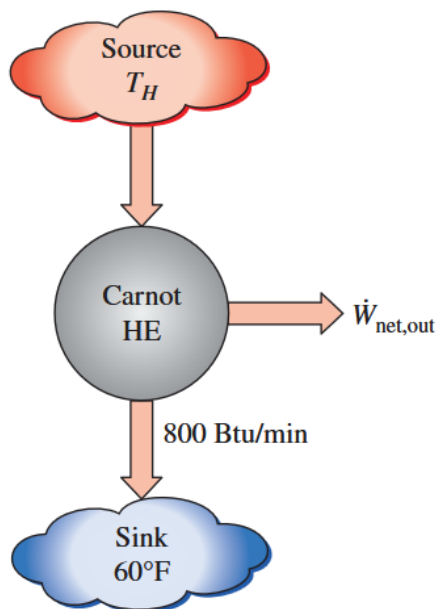


FIGURE P6-84E

Given:

- $\eta_{th} = 0.75$
- $\dot{Q}_L = 800 \text{ Btu/min}$
- $T_L = 60^\circ\text{F}$

Find:

- $\dot{W}_{net,out}$
- T_H

Given that this is a *carnot heat engine* (reversible and takes in heat to generate work), we can use the following relationships:

$$\eta_{carnot} = \frac{\dot{W}_{net,out}}{\dot{Q}_H} = 1 - \frac{\dot{Q}_L}{\dot{Q}_H} = 1 - \frac{T_L}{T_H}$$

Note: The heat terms may be expressed either as total heat (Q) or heat transfer rates ($\dot{Q}$), since the time dependence cancels in the efficiency ratio.

(a) By definition of thermal efficiency, we can find $\dot{Q}_H$:

$$\eta_{th} = 1 - \frac{\dot{Q}_L}{\dot{Q}_H}$$

$$\Rightarrow \dot{Q}_H = \frac{\dot{Q}_L}{1 - \eta_{th}} = \frac{[800 \text{ Btu/min}]}{1 - 0.75} = 3200 \text{ Btu/min}$$

Now we can find the net work output:

$$\eta_{\text{th}} = \frac{\dot{W}_{\text{net,out}}}{\dot{Q}_H} \implies \dot{W}_{\text{net,out}} = \eta_{\text{th}} \cdot \dot{Q}_H$$

$$\dot{W}_{\text{net,out}} = 0.75 \cdot [3200 \text{ Btu/min}] = 2400 \text{ Btu/min}$$

Converting to hp:

$$\dot{W}_{\text{net,out}} = [2400 \text{ Btu/min}] \cdot \left[\frac{778.169 \text{ ft} \cdot \text{lbf}}{1 \text{ Btu}} \right] \cdot \left[\frac{1 \text{ min}}{60 \text{ s}} \right] \cdot \left[\frac{1 \text{ hp}}{550 \text{ ft} \cdot \text{lbf/s}} \right] = \boxed{56.59 \text{ hp}}$$

(b) Using the Carnot efficiency relationship:

$$\eta_{\text{carnot}} = 1 - \frac{T_L}{T_H}$$

$$\implies T_H = \frac{T_L}{1 - \eta_{\text{carnot}}} = \frac{[60 + 459.67] \text{ R}}{1 - 0.75} = \boxed{2078.68 \text{ R}}$$

6-103

6-103 A Carnot refrigerator absorbs heat from a space at 15°C at a rate of $16,000 \text{ kJ/h}$ and rejects heat to a reservoir at 36°C . Determine the COP of the refrigerator, the power input, in kW, and the rate of heat rejected to high-temperature reservoir, in kJ/h.

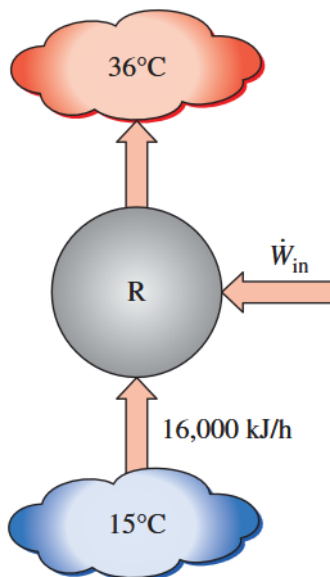


FIGURE P6-103

Given:

- $T_L = 15^\circ\text{C}$
- $\dot{Q}_L = 16,000 \text{ kJ/hr}$
- $T_H = 36^\circ\text{C}$

Find:

- β_{ref}
- $\dot{W}_{\text{in}}$ [kW]
- $\dot{Q}_H$ [kJ/hr]

First law for a closed system (cycle):

$$\dot{U} = \sum \dot{Q} - \sum \dot{W} \implies 0 = (\dot{Q}_L - \dot{Q}_H) - (-\dot{W}_{\text{in}}) \implies \dot{W}_{\text{in}} = \dot{Q}_H - \dot{Q}_L$$

Note: Sign conventions! Work is negative since it's entering and heat is positive when it's entering the system.

For a *carnot refrigerator* (reversible and uses work to absorb heat from the surroundings), we can use the following relationships:

$$\beta_{\text{carnot ref}} = \frac{\dot{Q}_L}{\dot{W}_{\text{in}}} = \frac{\dot{Q}_L}{\dot{Q}_H - \dot{Q}_L} = \frac{T_L}{T_H - T_L}$$

(a) Using the Carnot refrigerator relationship:

$$T_L = 15 + 273.15 = 288.15 \text{ K}, \quad T_H = 36 + 273.15 = 309.15 \text{ K}$$

$$\beta_{\text{ref}} = \frac{T_L}{T_H - T_L} = \frac{288.15}{309.15 - 288.15} = \boxed{13.72}$$

Note: CONVERT TO ABSOLUTE TEMPERATURES!

(b) Using relationship:

$$\beta_{\text{carnot ref}} = \frac{\dot{Q}_L}{\dot{W}_{\text{in}}} \implies \dot{W}_{\text{in}} = \frac{\dot{Q}_L}{\beta_{\text{carnot ref}}}$$

$$\dot{W}_{\text{in}} = \frac{[16,000 \text{ kJ/hr}]}{13.72} = 1166.18 \text{ kJ/hr}$$

Converting to kW:

$$\dot{W}_{\text{in}} = [1166.18 \text{ kJ/hr}] \cdot \left[\frac{1 \text{ hr}}{3600 \text{ s}} \right] = \boxed{0.323 \text{ kW}}$$

(c) Using the first law relationship:

$$\dot{Q}_H = \dot{Q}_L + \dot{W}_{\text{in}} = [16000 + 1166.18 \text{ kJ/hr}] = \boxed{17166.18 \text{ kJ/hr}}$$

6-106 **6-106** A heat pump is used to heat a house and maintain it at 24°C. On a winter day when the outdoor air temperature is −5°C, the house is estimated to lose heat at a rate of 80,000 kJ/h. Determine the minimum power required to operate this heat pump.

Given:

- $T_L = -5^\circ\text{C} = 268.15\text{ K}$
- $T_H = 24^\circ\text{C} = 297.15\text{ K}$
- $\dot{Q}_{\text{loss}} = 80,000\text{ kJ/hr}$

Find:

- *Minimum* power required to operate the heat pump

A heat pump uses work to transfer heat from a low temperature reservoir to a high temperature reservoir. Since the house stays at a constant temperature, we can see that

$$\dot{Q}_H = \dot{Q}_{\text{loss}}$$

Additionally, it asks for the *minimum* power required, and the most efficient process is the carnot process, so we can use the Carnot heat pump relationship:

$$\begin{aligned}\beta_{\text{carnot HP}} &= \frac{\dot{Q}_H}{\dot{W}_{\text{in}}} = \frac{T_H}{T_H - T_L} \\ \Rightarrow \beta_{\text{carnot HP}} &= \frac{[297.15\text{ K}]}{[297.15 - 268.15\text{ K}]} = 10.24 \\ \dot{W}_{\text{in}} &= \frac{\dot{Q}_H}{\beta_{\text{carnot HP}}} = \frac{\dot{Q}_{\text{loss}}}{\beta_{\text{carnot HP}}} \\ \Rightarrow \dot{W}_{\text{in}} &= \frac{[80,000\text{ kJ/hr}]}{10.24} = 7812.5\text{ kJ/hr}\end{aligned}$$

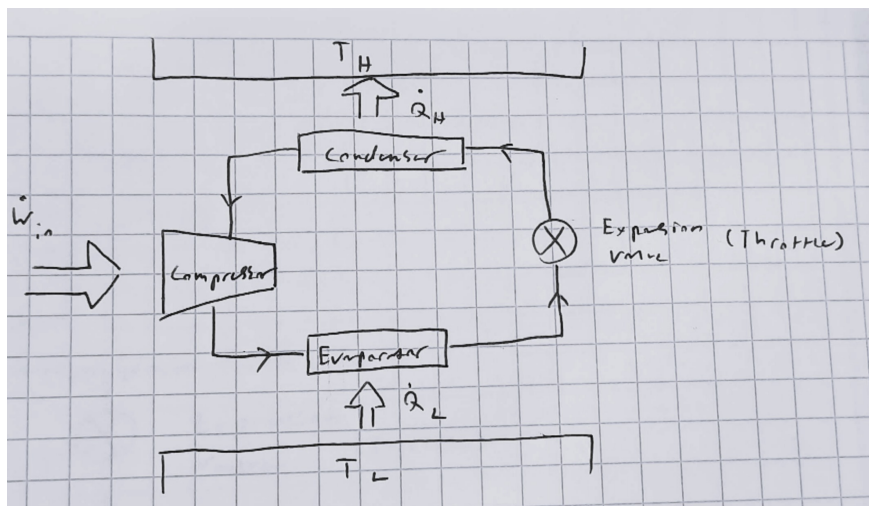
Converting to kW:

$$\dot{W}_{\text{in}} = [7812.5\text{ kJ/hr}] \cdot \left[\frac{1\text{ hr}}{3600\text{ s}} \right] = \boxed{2.17\text{ kW}}$$

- 6-135** **6-135** Consider a Carnot refrigeration cycle executed in a closed system in the saturated liquid–vapor mixture region using 0.96 kg of refrigerant-134a as the working fluid. It is known that the maximum absolute temperature in the cycle is 1.2 times the minimum absolute temperature, and the net work input to the cycle is 22 kJ. If the refrigerant changes from saturated vapor to saturated liquid during the heat rejection process, determine the minimum pressure in the cycle.

For saturated mixtures, pressure and temperature are always related. Additionally, as temperature increases, so does the saturation pressure. Since the R-134a remains as a saturated mixture throughout the cycle, the *minimum* pressure in the cycle MUST be the saturation pressure at the lowest temperature in the cycle. This is at the evaporator, where hot R-134a is cooled to T_L .

We are also told that the R-134a changes phases from saturated vapor to saturated liquid at the condenser, so we can use this to find Q_H (heat rejected as a result of phase change).



Note: Even though the R-134a needs to be colder than T_L to facilitate heat transfer, in an *ideal* reversible cycle, the R-134a is at the same temperature as the surroundings during heat transfer. Thus, we can use T_L to find the saturation pressure at the evaporator.

Given:

- $m = 0.96 \text{ kg}$
- T_L
- $T_H = 1.2T_L$
- $W_{\text{net, in}} = 22 \text{ kJ}$

Find:

- *Minimum* pressure in the cycle

(a) For a carnot refrigerator, we can use the following relationships:

$$\begin{aligned}\beta_{\text{carnot ref}} &= \frac{Q_L}{W_{\text{in}}} = \frac{Q_L}{Q_H - Q_L} = \frac{T_L}{T_H - T_L} \\ \implies \beta_{\text{carnot ref}} &= \frac{T_L}{1.2T_L - T_L} = \frac{T_L}{0.2T_L} = 5 \\ \implies Q_L &= 5 \cdot W_{\text{in}} = 5 \cdot [22 \text{ kJ}] = 110 \text{ kJ}\end{aligned}$$

Now we need to find Q_H to determine T_H . Using a first law analysis on the entire cycle (closed system):

$$\begin{aligned}W + Q_L &= Q_H \\ \implies Q_H &= W + Q_L = 22 \text{ kJ} + 110 \text{ kJ} = 132 \text{ kJ}\end{aligned}$$

(b) We can relate Q_H to the enthalpy change during the phase change at the condenser in order to find T_H . Using a first law analysis on the condenser (open system):

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left[h_i + \frac{1}{2}v_i^2 + gz_i \right] - \sum \dot{m}_e \left[h_e + \frac{1}{2}v_e^2 + gz_e \right]$$

- No work is done by the system ($\dot{W} = 0$)
- Steady state ($\frac{dE}{dt} = 0$)
- Neglect potential energy changes ($gz_i \approx gz_e$)
- Neglect kinetic energy changes ($\frac{1}{2}v_i^2 \approx \frac{1}{2}v_e^2$)
- By conservation of mass, $\dot{m}_1 = \dot{m}_2$

Thus, the first law simplifies to:

$$\begin{aligned}0 &= \dot{Q} + \dot{m}_1 h_1 - \dot{m}_2 h_2 \\ \implies \dot{Q}_H &= \dot{m}_1 (h_2 - h_1)\end{aligned}$$

Integrating over time:

$$Q_H = m(h_2 - h_1)$$

We know that the R-134a changes phases from saturated vapor to saturated liquid at the condenser, so

$$\begin{aligned}h_2 &= h_{f@T_H}, \quad h_1 = h_{g@T_H} \implies h_2 - h_1 = h_{fg@T_H} \\ \implies h_{fg@T_H} &= \frac{Q_H}{m} = \frac{[132 \text{ kJ}]}{[0.96 \text{ kg}]} = 137.5 \text{ kJ/kg}\end{aligned}$$

Finding the associated saturation temperature for this value of h_{fg} from the R-134a tables via linear interpolation using values at:

- $T = 60^\circ\text{C}$: $h_{fg} = 139.09 \text{ kJ/kg}$
- $T = 65^\circ\text{C}$: $h_{fg} = 132.05 \text{ kJ/kg}$

$$\begin{aligned}T - T_{60} &= \frac{T_{65} - T_{60}}{h_{fg@65} - h_{fg@60}} (h_{fg} - h_{fg@60}) \\ \implies T &= 60 - 0.710(h_{fg} - 139.09) \implies T_H = 61.129^\circ\text{C}\end{aligned}$$

Thus,

$$T_H = 334.279 \text{ K}$$

(c) Using the carnot refrigerator relationship to finally find T_L :

$$\beta_{\text{carnot ref}} = \frac{T_L}{T_H - T_L}$$

$$\Rightarrow T_L = \frac{\beta_{\text{carnot ref}} \cdot T_H}{1 + \beta_{\text{carnot ref}}} = \frac{5 \cdot [334.279 \text{ K}]}{1 + 5} = 278.57 \text{ K}$$

Convert to Celsius:

$$T_L = 278.57 - 273.15 = 5.415^\circ\text{C}$$

(d) Finally, we just need to find the saturation pressure at T_L to find the minimum pressure in the cycle. Linear interpolation using values at:

- $T = 4^\circ\text{C}$: $P_{\text{sat}} = 337.90 \text{ kPa}$
- $T = 6^\circ\text{C}$: $P_{\text{sat}} = 362.23 \text{ kPa}$

$$P - P_4 = \frac{P_6 - P_4}{T_6 - T_4}(T - T_4)$$

$$\Rightarrow P = 337.90 + 12.165(T - 4) \Rightarrow P_{@T_L} = 350.065 \text{ kPa}$$

$$\Rightarrow \boxed{P_{\text{min}} = P_{@T_L} = 350.065 \text{ kPa}}$$

6-159 **6-159** Consider a Carnot refrigerator and a Carnot heat pump operating between the same two thermal energy reservoirs. If the COP of the refrigerator is 3.4, the COP of the heat pump is
 (a) 1.7 (b) 2.4 (c) 3.4 (d) 4.4 (e) 5.0

Looking at the carnot efficiency relationship for a refrigerator:

$$\beta_{\text{carnot ref}} = \frac{T_L}{T_H - T_L} = 3.4$$

For a heat pump, we have:

$$\beta_{\text{carnot heat pump}} = \frac{T_H}{T_H - T_L}$$

Since $T_H = (T_H - T_L) + T_L$, we can rewrite the heat pump relationship as:

$$\beta_{\text{carnot heat pump}} = \frac{(T_H - T_L) + T_L}{T_H - T_L} = 1 + \frac{T_L}{T_H - T_L} = 1 + \beta_{\text{carnot ref}}$$

$$\Rightarrow \beta_{\text{carnot heat pump}} = 1 + 3.4 = \boxed{4.4 \text{ (d)}}$$